RaktSetu — Solution brief

Predictive blood and platelet sharing between Jaipur's blood banks · COMMIT FOR GOOD 2026 · MIT licence

1. Problem

Blood-bank coordinators in Jaipur cannot move stock to where it will be needed before it runs out or expires, because each bank plans alone and e-RaktKosh only shows today's stock, which causes platelet shortages in dengue season while units are discarded unused a few kilometres away.

Who is affected. Patients who need red cells (surgery, trauma, obstetric haemorrhage, thalassaemia) and platelets (dengue, chemotherapy). Their families, who are sent from bank to bank. Blood-bank officers. Donors whose blood expires unused.

Evidence.

- National collection reached 14.6 million units in 2024–25, roughly the requirement, so the problem is distribution, not supply (ORF, 2025).
- 5–15% of collected units are wasted (ORF, 2025). About 28 lakh units were discarded between 2012 and 2016 (NACO via RTI).
- In eight states including Rajasthan, median stock is 0.6 units per 1,000 people and 22% of banks rarely update e-RaktKosh (BMJ Global Health, 2024).
- Platelets last 5 days. Rajasthan had 12,514 dengue cases in 2024 and 20,749 in 2021 (NCVBDC). One Jaipur bank's platelet requests went from 15–20 a month to 50–60 a day in an October surge.

What exists, and the gap. e-RaktKosh provides visibility. Phone-around sharing happens after a patient is already waiting, and nobody moves near-expiry units proactively. Nothing forecasts, recommends or warns.

2. Approach

Input: a date (plus, optionally, real stock typed in, vans available, or a crisis scenario). Output: today's van trips with reasons, shortfalls escalated to donor drives, and a 4-week platelet surge warning.

Question	Method	Baseline it must beat
How many requests at each bank, 1–5 days ahead?	Gradient boosting (Poisson), features: recent requests + lagged real rainfall	7-day average; same weekday last week
What should move today, given limited vans?	Integer programme (OR-Tools CP-SAT): maximise shortages + expiries averted − travel; vans, cooler boxes, each bank's own protected need	Phone-around after a stock-out
Will the city run short of platelets in 2–4 weeks?	City-level gradient boosting on rainfall lagged 4–12 weeks	Current level persists; same week last year
What happens in a crisis?	Day-by-day inventory simulator: ageing, expiry, ABO/Rh substitution, phone-around	—

Data. Real: 18 blood banks (OpenStreetMap), road times (OSRM), Jaipur weather 2016–2026 (Open-Meteo), Rajasthan dengue cases (NCVBDC), blood-group mix (10,000-donor study). Generated (seeded, every assumption listed): daily requests and donations per bank. Results validate the method, not a real-world outcome.

3. Results (simulation, 5 seeds)

	2024 dengue season	2025 full year
Unmet requests vs phone-around	**−36%**	**−80%**
Expired units vs phone-around	**−52%**	**−43%**
Red-cell fill rate	96.9% → **99.8%**	93.9% → **99.8%**
Red-cell wastage	10.3% → **4.8%**	11.6% → **5.5%**
Cost	+2,650 van-km	+11,300 van-km

- **Sensitivity:** the improvement holds for every assumption varied (dengue peak 1.5–4×, collection 0.98–1.18× demand, 1–3 vans, service level 0.7–0.9). **One van is enough** for Jaipur.
- **Forecasting:** the ML model beats the 7-day average (WAPE 33.2% vs 35.8% red cells; 52.8% vs 55.6% platelets; noise floor 33.1% / 51.5%). Most of the gain comes from coordination, not the forecast.
- **Surge radar:** 3–4 weeks ahead, error 13.6–14.6% vs 20.5–24.8% for persistence. For next week, persistence is as good.

4. What we learned, and what is next

1. **Coordination beats prediction.** A simple forecast plus an optimiser captures most of the value. That makes the system cheap to deploy.
2. **At the dengue peak the city is genuinely short of platelets.** No reallocation fixes that, so the 2–4-week surge radar, which triggers donor drives, is the right second lever.
3. **Pilot plan:** one district, read-only e-RaktKosh stock export, plan reviewed by the District Blood Transfusion Officer each morning for 8 weeks of dengue season, then compare discards and turn-aways against the previous season.

Limitations: synthetic demand; bank profiles guessed; platelets pooled across groups; inter-bank transfer rules to be confirmed with the Rajasthan State Blood Transfusion Council; more driving.

Repository: README with install and run steps, `data/ASSUMPTIONS.md`, `docs/ARCHITECTURE.md`, `docs/EVALUATION.md`, 15 tests, Dockerfile, CONTRIBUTING, MIT licence.